AUTOMOTIVE OTA

PERSONAL PROJECT

07-11-2025

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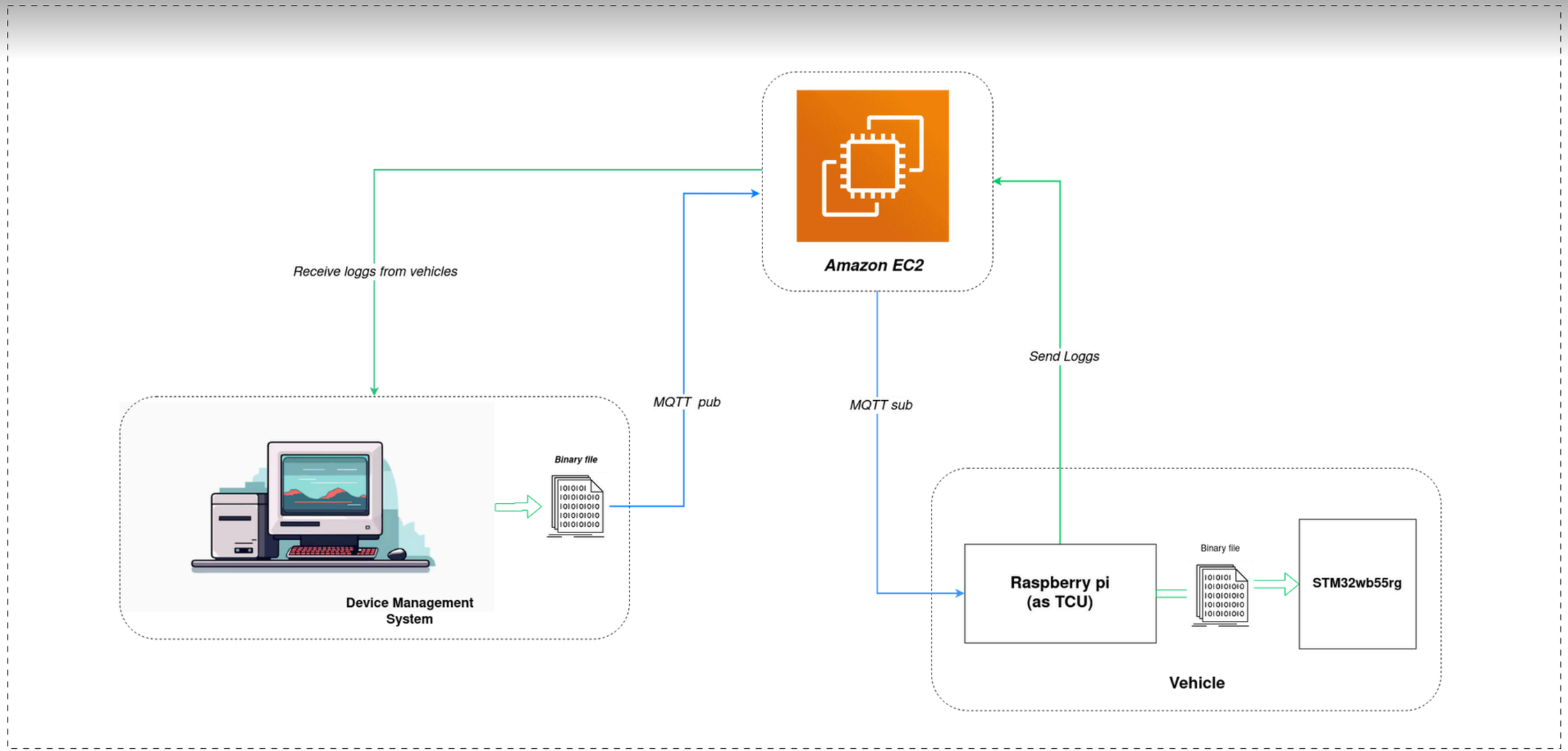
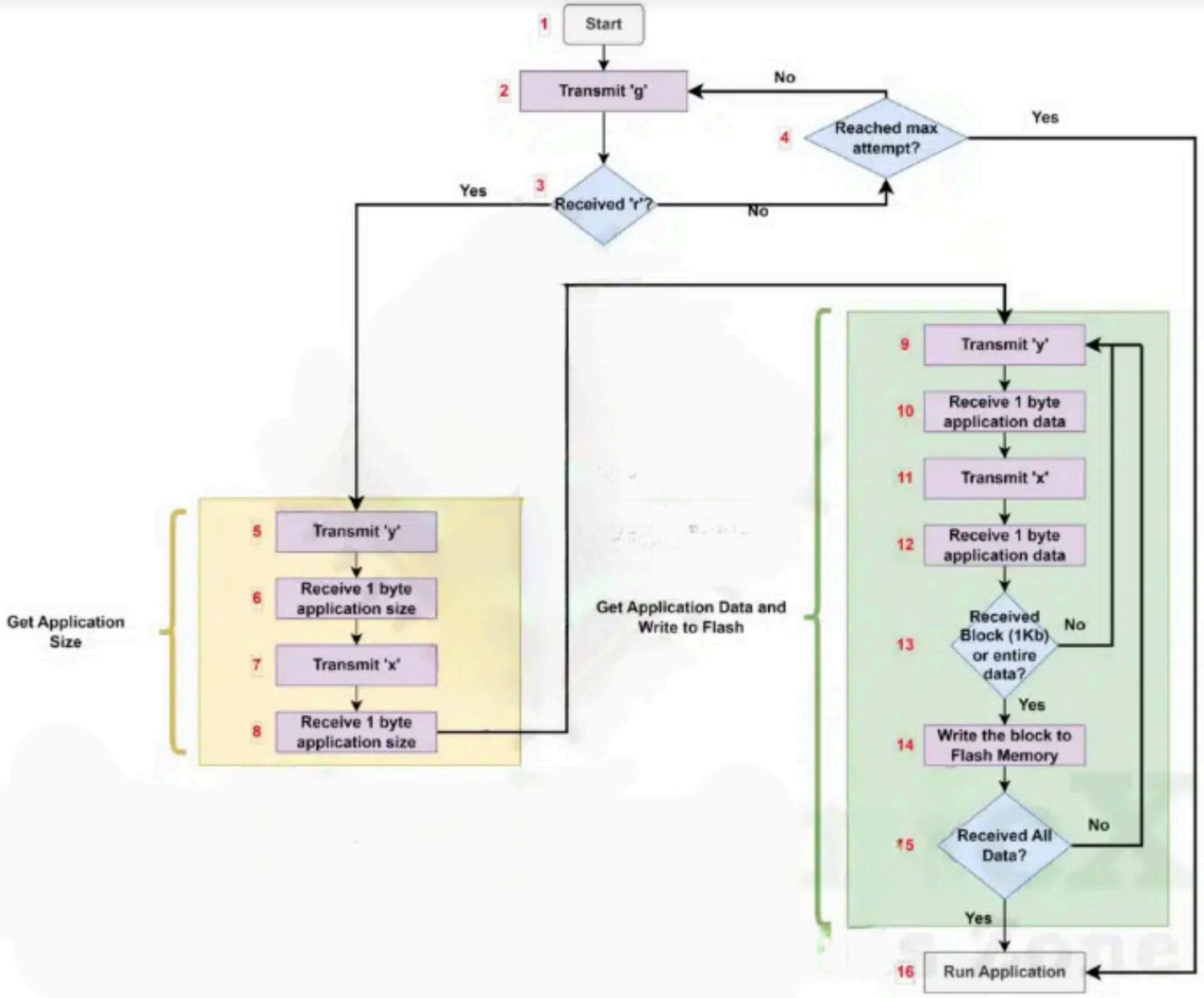
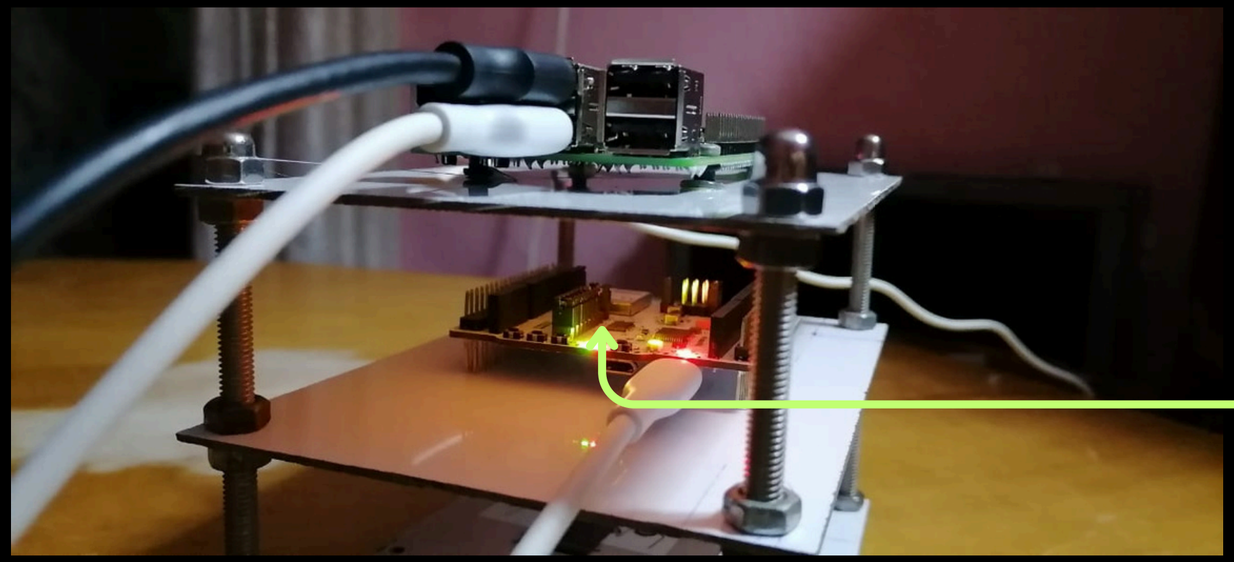
GENERAL ARCHITECTURE



UPADATE WORKFLOW

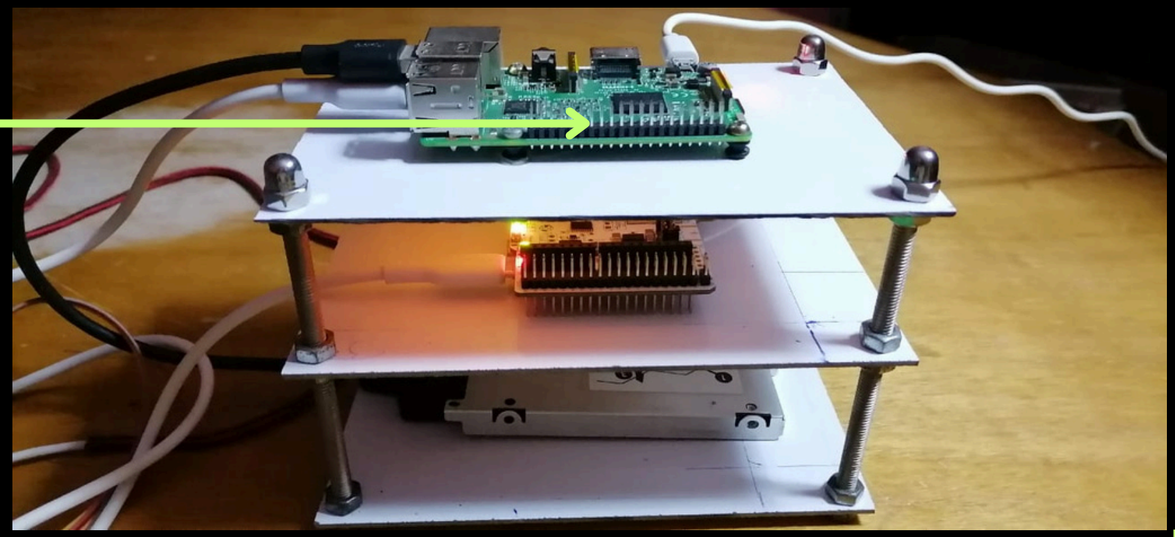


PROTOTYPE SETUP



STM32WB55RG

Raspberry pi 3



RESULTS

SEND THE UPDATE FROM THE DEVICE MANAGEMENT SYSTEM TO THE VEHICLES VIA A BROKER ON THE CLOUD

```
elhaloua@192:~/STM32CubeIDE/workspace_1.19.0/09_2025_CUSTOM_BL/OTA_EV_update/OTA_EV_
Sender/build$ ./OTA_EV_Send
Firmware sent successfully!
elhaloua@192:~/STM32CubeIDE/workspace_1.19.0/09_2025_CUSTOM_BL/OTA_EV_update/OTA_EV_
Sender/build$
```

Send the encoded update the file to the broker on AWS 2CE using mqtt

Receive the new firmware from the broker on raspberry pi.

```
┌──(greencrow㉿greencrow)-[~]
└─$ mosquitto_sub -h 16.170.237.131 -p 1883 -t "ota/firmware" -C 1 > /tmp/recieved_firmware.b64

┌──(greencrow㉿greencrow)-[~]
└─$ ls /tmp/recieved_firmware.b64
/tmp/recieved_firmware.b64

┌──(greencrow㉿greencrow)-[~]
└─$ cat /tmp/recieved_firmware.b64
Salaamo Alaykom
AAADIJULAQgpCQEIMQkBCDkJAQhBCQEISQkBCAAAAAAAAAAAAAAAAAAAAABRCQEIXwkBCAAAAABt
CQEIewkBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0L
AQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsB
CO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI
7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjt
CwEI7QsBCO0LAQjtCwEI7QsBCO0LAQjtCwEI7QsBCAAAAAAQtQVMI3gzuQRLE7EESK/zAIABIyNw
```

TRANSFER THE UPDATE VIA UART

```
elhaloua@192:~/STM32CubeIDE/workspace_1.19.0/09_2025_CUSTOM_BL/UART_update_sender/PcTool$ ./etx_ota_app 25 nucleo_wb55_application.bin
Opening Port 25...
Opening Binary file : nucleo_wb55_application.bin
File size = 19916
Waiting for OTA Start
Sending r...
Sending FW Size...
Sending Data...
Block 1 Transmitted...
Block 2 Transmitted...
Block 3 Transmitted...
Block 4 Transmitted...
Block 5 Transmitted...
Final Block 5 Transmitted...
elhaloua@192:~/STM32CubeIDE/workspace_1.19.0/09_2025_CUSTOM_BL/UART_update_sender/PcTool$
```

Once the raspberry pi receives the binary file from the broker it transfer it to the stm32 via uart

START THE CUSTOM BOOTLOADER

WHEN NO VALID APPLICATION IS FOUND, AND NO DATA RECEIVED FROM THE HOST.

```
Starting custom Bootloader(2.0)
4
3
2
1
0
gggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggggNo Data Received for Firmware Update
No valid application found, staying in bootloader
```

WHEN A VALID APPLICATION IS FOUND

```
Custom Starting Bootloader(0.1)
Gonna Jump to Application
                              Starting Application(0.1)
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
Salaam from the App.
```

THE CUSTOM BOOTLOADER RUNNING

Once the stm32 is powered on the custom BL starts running, which will print a message blinking a LED indicating that is has started, before it move on to the firmware update function, where it will continuasly send a command’g’ via UART to the TCU (in our case it is the RPI), if it receives ‘r’ from the TCU the new firmware transmission will started.

If no respense from the TCU is received the BL automaticly jumps to the old application.


THANK YOU